

Answer any five of the following questions.
All parts of a particular question must be answered consecutively.

1. Consider an economy whose production function takes the form $Y = A_t K^\alpha L^{1-\alpha}$, where K_t and L_t denote the amount of capital and labor employed respectively and $0 < \alpha < 1$. Suppose that the population (L) and productivity (A) grow at exogenous n and g rates respectively. The economy is closed and without government. Households save a fixed percent of their income and capital stock depreciate at δ rate.

- (a) Assume that $g = 0$. Derive the fundamental equation of the Solow model and calculate the growth rates of y , k , and c in the steady state. [4]
- (b) Find closed form expressions for y , k , and c . Show the steady state situation in a graph. [2]
- (c) Now assume that $g > 0$. Derive the condition for steady state and calculate growth rates of y , k , and c . [4]
- (d) List the stylized facts about growth. Evaluate the performance of the Solow model with regard to the stylized facts. [4]

2. Consider a Ramsey-Cass-Koopmans model with the household's preference given by

$$\sum_{t=0}^{\infty} \beta^t \log c_t$$

The household may spend any fraction of their unit time endowment to work. And the production technology is given by

$$y_t = k_t^\alpha l_t^{1-\alpha}$$

where y_t satisfies usual assumptions. Output can be used either for consumption or investment. The law of motion of capital is

$$k_{t+1} = i_t + (1 - \delta)k_t$$

where $0 < \delta < 1$ is the rate of depreciation. The initial capital stock is k_0 .

- (a) Set up the social planner's problem for this economy. Mention state and control variables. [2]
- (b) Use the Lagrangian method (Kuhn-Tucker conditions) and the dynamic programming method to solve this problem and compare the results. [8]
- (c) Find the optimal policy function $k_{t+1} = g(k_t)$. [4]

3. (a) Consider the Lucas's one sector model of endogenous growth. In addition to physical capital, households can accumulate human capital. Household's utility function is given by

$$u(c) = \frac{c^{1-\theta} - 1}{1-\theta}$$

Solve the household's problem and derive the growth factor of C as function of parameters only. [6]

(b) Show that along BGP, C , Y , H , K , and I grow at the same rate. [3]

(c) Suppose that there are two sectors in the economy. One sector (called C firm) uses capital and human to produce consumption and capital investment good, another sector (called H firm) uses only human capital to produce human capital investment. $\phi_t \in [0, 1]$ is the fraction of human capital used in the C firm. Set up the social planner's problem for this economy and show that along BGP, C , Y , H , K and I grow at the same rate. [5]

4. Suppose that the representative firm produces output given by the production function $y_t = a k_t^\alpha$. This infinitely lived firm owns capital stock and takes investment decision to maximize lifetime profit. Assume that the price of output, P , is one and capital depreciates at the rate δ .

A}'(K) -

(a) Construct the firm's problem. Find and interpret the user cost of capital. [3]

(b) What is the limitation of the model you designed in part (a)? How can it be fixed? [1]

(c) Suppose that the firm faces a quadratic adjustment cost, given by

$$C(i_t, k_t) = \frac{i_t^2}{2\phi k_t}$$

while making investment decision. Derive and interpret Tobin's q and find an implicit investment schedule. [5]

(d) From the optimality conditions in part (c), find a system of two differential equations in k and q . [2]

(e) Linearize the system you found in part (d) and discuss the stability issues of the system. [3]

5. Consider a finitely-lived representative consumer who inelastically supplies one unit of labor. This consumer-worker receives wage w for per unit of labor supplied and gets interest r from saving. The problem to this consumer is to choose how much to save and consume over her lifetime. The economy is closed without government and population does not grow. Now answer the following questions.

(a) Set up the representative consumer's problem and explain it. [2]

(b) Find the Euler equation and interpret it. [4]

(c) Find a closed-form solution of consumption and show the pattern of bond holding with constant labor income. [6]

(d) Show that the model delivers properties of life-cycle hypothesis when labor-income is non-constant. [4]

6. (a) Derive the IS and LM curves for a small open economy. Using reconstructed IS-LM model, discuss the effectiveness of fiscal and monetary policy under flexible exchange rate with perfect and imperfect capital mobility. [9]

(b) Prove that nominal rigidity can produce non-vertical Aggregate Supply Curve (ASC) but does not guarantee pro-cyclical real wage. Discuss in light with both sticky price and sticky wage models. [5]

7. Consider the infinitely lived representative household that derives utility from both consumption (c_t) and leisure ($1 - l_t$). The production function of the representative firm is given by

$$f(k_t, l_t, z_t) = z_t k_t^\alpha l_t^{1-\alpha}$$

where z_t is the productivity parameter. Assume that z_t follows AR(1) process i.e., $z_{t+1} = \phi z_t + \varepsilon_{t+1}$, where ε_t is a white noise error term.

(a) Construct the Household's problem and show that MRS between consumption and leisure is the real wage. [2]

(b) Assuming log-utility show that short run labour supply curve is upward sloping but labor supply is independent of wage in the long run. [5]

(c) Assuming CRRA utility $u(c, h) = \frac{c^{1-\theta}}{1-\theta} - \frac{h^{1-\theta}}{1-\theta}$, find the elasticity of labor supply. [3]

(d) Mention two important limitations of early business cycle models. Discuss Hansen's indivisible labor supply solution as an attempt to overcome these limitations. [4]

8. (a) What are the basic tenets of New Classical Economics? How does this approach differ from Classical and Keynesian approaches? [3]

(b) How does the concept of rational expectations revolutionize the way we think about macroeconomic dynamics? Discuss. [4]

(c) Construct a simple macroeconomic model with the assumption of rational expectations and show that systematic policy becomes ineffective. [7]

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2020
Course Econ 503: Development Economics

Time: 4 hours

Full Marks: 70

Instructions

- Answer any 05 (Five) questions
- Marks on the right margin

[7+7=14]

1. Make an account of the investment strategy as sketched in the First Five Year Plan of Bangladesh.

2. Can we explain the transition of this strategy over five decades since the country's independence?

[14]

Does economic growth guarantee an increase in life expectancy? Explain with different experiences from different parts of the world.

[14]

Although the contribution to GDP of the agriculture sector has gone down as a stylized fact of development, the sector has undergone transition and still contributes to the economy of Bangladesh in a major way—Explain.

[7+7=14]

- 4.
- a. What is meant by “neoclassical counterrevolution”. What are its principal arguments? How valid do you think they are? Explain.
 - b. Is the neoclassical, free market theory compatible with dependence theory? How may these two approaches work together? Explain.

[4+5+5=14]

- 5.
- a. How does the ‘coordination failure approach’ explain sustained underdevelopment?
 - b. Explain a diagrammatic approach to illustrate multiple equilibria with possible coordination failure.
 - c. What is a “middle income trap”? How is the concept relevant to the current economic development of Bangladesh?

[4+5+5=14]

6. Explain the concept of “sustainable net national income”.

Is it possible to achieve growth without further damage to the environment?

Explain the policy relevance of the Kutznet’s curve in light of local level pollution as well as public goods like greenhouse emission

[6+8=14]

- 7.
- a. Explain the relation between disaster and sectoral and national GDP as explained by Benson and Clay (2004).
 - b. Explain the basic principles of the Paris Agreement in connection to economic growth.

[4+10
12+8+4=14]

8. How does the concept of SDG challenge other economic development pathways?

How can SDG1 be related to secure global food security? Explain.

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2020
Course Econ 512: Political Economy of Globalization

Time: 4 hours

Full Marks: 70

Instructions

- Answer any 05 (Five) questions
- Marks on the right margin

Q1.

[8+6=14] ✓

বিশ্বায়ন সম্পর্কিত কয়েকটি প্রধান তত্ত্ব সময়ে উপস্থিত কর। রোজা লুক্সেমবুর্গের তত্ত্ব কি বর্তমান স্থায়ী যুদ্ধাবস্থা ব্যাখ্যা করতে পারে? আলোচনা কর।

Briefly present some major theories of globalization. Can the theory of Rosa Luxembourg explain the present day permanent war situation? Discuss.

Q2.

[6+8=14] ✓

চতুর্থ শিল্প বিপ্লব বলতে কি বোঝায়? শোশানা যুবফ কীভাবে 'নজরদারী পুঁজিবাদ' ব্যাখ্যা করেছেন? আলোচনা কর।

What is meant by the Fourth Industrial Revolution? How did Shoshana Zuboff explain 'surveillance capitalism'? Discuss.

Q3.

[6+8=14] ✓

কীভাবে বিশ্বায়নের পেছনে চালিকা শক্তি হিসাবে ব্রেটন উডস ইন্সটিটিউটগুলো বিকশিত হয়েছে? বিশ্ব প্রবৃদ্ধির সাথে বৈশ্বিক বৈষম্য বৃদ্ধি এবং জলবায়ু পরিবর্তন সমস্যা তুমি কীভাবে ব্যাখ্যা করবে?

How did the Bretton Woods Institutes emerge as a driving force behind globalization? How do you explain the rise of global inequality and climate change problems with global growth?

Q4.

[6+8=14]

বিশ্ব শ্রমশক্তির নারীকরন তুমি কীভাবে ব্যাখ্যা করবে? বিশ্ব অর্থনীতিতে প্রবাসী শ্রমিকের ক্রম বর্ধমান ভূমিকার কারণ ও তাৎপর্য আলোচনা কর।

How do you explain the feminization of the global workforce? Discuss the causes and significance of the increasing role of migrant labour in the global economy.

P.T.O.

Q5.

[4+10=14]

বিশ্বায়ন প্রক্রিয়ার পেছনে কার্যকর প্রধান শক্তিগুলো কী কী? আলানি অথবা কৃষিতে বিশেষ গুরুত্ব দিয়ে বিশ্ব অর্থনীতিতে বহুজাতিক সংস্থাসমূহের কার্যক্রম এবং প্রভাব পরীক্ষা কর।

Who are the major effective actors behind the globalization process? Examine actions and influences of Multi-National Corporations (MNCs) on the global economy with special reference to energy or agriculture.

Q6.

[4+10=14] ✓

ওয়াশিংটন কনসেন্সাস বলতে কী বোঝায়? নয়া উদারনীতিবাদ কীভাবে উন্নয়নশীল দেশসমূহের সরকারগুলোকে বিশ্বায়নের সাথে সংযুক্ত হতে প্রভাবিত করে?

What is the Washington Consensus? How does neoliberalism influence the governments of developing countries toward integrating globalization?

Q7.

[14]

বিশ্বায়ন প্রক্রিয়া কী মনুষ্য অভিপ্রয়ানকে উন্নয়নশীল থেকে ধনী দেশে উৎসাহিত করে? এ ব্যাপারে তোমার মতামত ব্যাখ্যা কর।

Do you think that the globalization process approves human migration from developing countries to richer countries? Explain your views?

Q8.

[7+7=14]

নীচের যে কোন দুটির উপর টীকা লিখ

ক) নিওক্লাসিকাল নির্ভরশীলতা তত্ত্ব।

খ) বর্তমান বিশ্বব্যবস্থায় চীন।

গ) নয়া উদারনৈতিক অর্থ ব্যবস্থা।

ঘ) টেকসই উন্নয়ন লক্ষ্যমাত্রা।

Write Short Notes ON ANY TWO of the following:

a. Neoclassical dependency theory

✓ The role of China in today's globalization

c. Neoliberal financial system

✓ Sustainable development goals (SDGs)

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2020

Course: Econ 501: Advanced Microeconomic Theory

Time: 4 hours

Full Marks: 70

Instructions

- Answer any 05 (Five) questions by taking at least one question from each section
- Marks on the right margin

SECTION A:

Q1. [5+9=14]

- a) In what respect does general equilibrium differ from partial equilibrium? Examine with an example.
- b) Graphically explain the problems of existence, uniqueness and stability in connection with general equilibrium.

Q2. [8+(2+2+2)=14]

- a) 'If externality is present, it would lead to a socially inefficient production' Explain with an appropriate example.
- b) Suppose the cost functions of firm A and firm B are:
 $C_A = 0.5 q_A^2 + 8q_A - 0.5q_B^2$; and $C_B = 0.4q_B^2 + 10q_B + 0.025q_A^2$.
 - i) Find the outputs and profits of both firms when they operate separately.
 - ii) Find the outputs and profits of both firms when they operate jointly.
 - iii) Which one is the socially efficient production plan and why?

Q3. [8+6=14]

- a) Graphically explain how an external shock in the product market affects the factor market and the mechanism through which equilibrium is restored in both markets.
- b) Explain the consumption equilibrium in a simple general equilibrium model through an Edgeworth box diagram.

SECTION B:

[6+8=14]

- a) Explain the underlying assumptions of marginal utility of money that makes the Hicks-Kaldor compensation, correct?
- b) Derive the grand utility possibility frontier and determine the 'point of bliss'. Does such point ensure maximum social welfare? Justify.

[6+4+4=14]

- a) Define social welfare function. How is it used as a criterion to measure social welfare? Explain.
- b) Critically explain the three key factors that Pigou dealt with in analyzing social welfare.
- c) Does perfect competition ensure welfare maximization? Examine.

SECTION C:

Q6.

[6+8=14]

- What are the informational asymmetries in market models? Examine with examples.
- Suppose each entry-level software programmer in Palo Alto, California, has either high or low ability. All potential employers value a high-ability worker at \$12,000 per month and a low-ability worker at \$6,000. The supply of high-ability workers is $Q_H = 0.05(W - 2,000)$ and the supply of low-ability workers is $Q_L = 0.1(W - 2,000)$, where W is the monthly wage. If workers' abilities are observable to employers, what are the equilibrium wages? How many workers of each type do employers hire? If workers' abilities are not observed by employers, what is the equilibrium wage? How many workers of each type do employers hire? What is the deadweight loss due to asymmetric information? Graphically present your findings.

[5+9=14]

- What does a labor market unravel? Briefly discuss.
- What do you mean by separating and pooling equilibrium? Suppose the value of a worker's marginal product is \$55 per hour for high-ability workers and \$15 per hour for low-ability workers. The preferences of high-ability workers correspond to the utility function $U_H(E, W) = W - 4E$, and the preferences of low-ability workers correspond to the utility function $U_L(E, W) = W - 8E$. By law, everyone is required to attend at least 10 years of school. One-quarter of the population has low ability, and three-quarters have high ability. What can you say about the educational attainment of each type of worker in a separating equilibrium, and in a pooling equilibrium? Are workers better off with the most efficient separating equilibrium, or with the most efficient pooling equilibrium?

Q8.

[5+5+4=14]

- In Bangladesh, bus drivers have at times been paid based on the number of riders they pick up. Discuss the likely differences in driver behavior under this plan compared to a plan in which drivers are paid by the day.
- What do you mean by an incentive scheme? Suppose a salesperson works for a car dealership for 40 hours per week, but may not choose to work hard all of the time. The dealership's owner cannot observe the salesperson's effort, but can observe the number of cars sold. The salesperson's personal cost of working at the dealership is $C = 1,000 + H^2$, where H is the number of hours during which he works hard. The corresponding marginal cost is $MC = 2H$. Without any effort, the salesperson will, on average, generate a profit of \$800. With each hour of high effort, he has a 10 percent chance of selling a car. Each car sale generates a profit of \$1,000. What is the efficient number of hours of high effort? How much surplus does the relationship generate?
- Suppose the owner gives the salesperson \$1,200 in base pay plus a bonus of \$500 for each car he sells. How hard will the salesperson work? Graphically describe an incentive scheme that leads to the efficient effort level and allows the owner to keep all of the surplus.

Question for MSC Tutorial 503

Answer any two:

1. How does environmental consideration relevant in economic development? Do you think Kuznet's curve can be flattened?
2. What is a Comprador Group? What economic school of thought explains the reference of this group to explain a country's development process.
3. How does adaptation relate to economic decisions? Explain the basis principles of Paris Agreement.
4. How does the concept of SDG challenge other economic development pathways so forth? How does SDG 1 can be related to secure global food security?

1st Tutorial

M.Sc; Course: 503 (Development economics)

Date 17.07.2022

1. Make an account of the investment strategy as sketched in Bangladesh's First Five Year Plan. Can we explain the transition of this strategy over Five decades since the Country's Independence?
2. Does Economic Growth raise life Expectancy? Please explain.
3. Though the contribution to GDP of agriculture sector has gone down as a stylized fact of development, the sector has undergone through transition and still contributes to the economy of Bangladesh in significant ways –please explain.
4. Explain the core components of different growth theories (school of thought).

T-1

1. Economic growth theory
- ✓ 2. Agriculture Sector in BD - Slide ✓
- ✓ 3. Preston Curve - with example ✓
- ✓ 4. First Five Year Plan - Investment plan, GDP growth, labor intensive (Slide) ✓

T-2

- ✓ 1. Underdevelopment as co-ordination failure (4.1) ✓
2. Samir Amin [Todaro-3] Dependency Theory
3. Environment and Development (10 - Todaro)
- ✓ 4. SDG indicator and Target ✓
5. Climate change
- ✓ 6. Paris Agreement ✓

Todaro - 4
4.1 ; 4.2

1. Neo-Colonial Development (Todaro) (122)
2. Vicious circle of poverty
3. Middle income trap (69) ✓
4. Kuznets Curve - (Todaro) (5.2 - 224) ✓
5. Bension Theory in Climate change

4.1, 4.2
full < 10

1. Agriculture Sector
2. Preston Curve
3. SDG
4. Climate change
5. Paris agreement
6. Kuznets Curve
7. 4.1 + 4.2
8. Dependency Theory (Samir Amin)
9. Middle income Trap
10. Neo-Colonial Development
11. Vicious circle of poverty
12. First five Year Plan
13. Todaro - 10
14. Benson Theory in EC

~~SDG~~

~~Preston~~

~~Environment + Kuznet + Paris~~

~~Growth theory~~

~~Agricultural sector~~

~~first 5 year Plan~~

AKTER
Eco-45

Department of Economics
Jahangirnagar University
Course No-512. Political Economy of Globalization
Final Tutorial Exam. June 15, 2022
Full Marks: 20. Time : 45 Minutes
Answer any ONE

1. How do you conceptualize globalization? Explain the concept of TCC and the World System Theory. (8+12)
2. How do 'surveillance capitalism' and the war machine shape the existing global order? Explain. (20)
3. Identify the major actors in global capitalism. How do you read the danger of climate change and the effectiveness of global response? (8+12)
4. Write short notes on (any two): (a) Russia-Ukraine war (b) Global Refugee Crisis (c) Fourth Industrial Revolution (d) US- China tension in global system. (10+10)

1. Theories of Globalization (Imperialism)
2. UCR
3. Role of Global institutions in Globalization
4. Contemporary Issue (Russia-Ukraine, Sri-lanka, USA-China)
5. Driving force → MNCs, UN system
6. Migrant labor, Surveillance system, Refugee crisis
7. Global system in BD context
 - a) International trade (Global Economy)
 - b) Remittance / Migrant labor (80s)
 - c) Foreign Aid / FDI
8. Internet ; Mobile ; Telecommunication ; Online Browsing ; Mobile Banking ; Satellite
9. MDG, SDG, Human Rights, CEDAW, UNCLOS

Theories of Globalization and Liberalisation

Tutorial Examination

MSc. Program, Department of Economics, Jahangirnagar University

Course: Econ-502 (Advanced Macroeconomic Theory)

Time: 60 minutes; Full Marks: 20

Answer the following question.

T-2

502

1. Consider a finitely-lived representative consumer who inelastically supplies one unit of labor. This consumer-worker receives wage w for per unit of labor supplied and gets interest r from saving. The problem to this consumer is to choose how much to save and consume over her lifetime. The economy is closed without government and population does not grow. Now answer the following questions.
- (a) Set up the representative consumer's problem and explain it. [2]
 - (b) Set set up the Hamiltonian and find the optimally conditions. [2]
 - (c) Find the Euler equation and interpret it. [3]
 - (d) Find a closed-form solution of consummation and show pattern of bond holding with constant labor income. [6]
 - (e) Show that the model delivers properties of life-cycle hypothesis when labor-income is non-constant. [4]
 - (f) Suppose that utility function takes the form $U(c_t) = \frac{C_t^{1-\sigma} - 1}{1-\sigma}$. Find the Euler equation. [3]

OR

2. If you find question 1 extremely challenging, you may also write a short thesis on consumption theory from your lifetime learning. In that case, you can get no higher than 10 mark. [10]

MSc. Program, Department of Economics, Jahangirnagar University

Course: Econ-502 (Advanced Macroeconomic Theory)

Time: 60 minutes; Full Marks: 20

Answer any one of the following questions.

1. (a) Construct a simple Solow model with fixed technology and population growth, and show that per capita variables grow at zero rate in the long run.
 (b) Suppose that the production function takes form $Y = Ak^\beta L^{1-\beta}$. Find the steady state values of k, y , and c .
 (c) Assume that technology grows at m rate and capital depreciates at δ rate. Show that per capita variables grow at m rate. Find steady state value of k in this case.

2. Consider a representative household that inelastically supplies one unit of labor. This consumer-worker receives wage w for per unit of labor supplied. In addition to consumption and bond holding, this household also holds money, which increases utility because of its liquidity services. Her problem is to choose how much to consume, save, work, and hold money so as to maximize her lifetime utility. The economy is closed without government and population does not grow. Now answer the following questions.
 (a) Set up the representative household's problem and develop the demand side, summarized by the IS-LM model, for this economy.
 (b) Suppose that the central bank conducts monetary policy according to some interest rate rule, and the ASC is upward sloping. Using these facts summarize the New Keynesian AD-AS model.
 (c) Suppose that the economy is hit by a positive demand shock. Show that this demand shock has real effects in the short run. Feel free to use additional assumptions to simply and solve the model analytically.

Answer any two questions taking ONE from each section

Section A: Welfare Economics

1. a) Define social welfare. Critically explain the following criteria as a measure of social welfare:
 - i) Growth of GNP
 - ii) Bentham's Criteria
 - iii) Cardinalist approach.
- b) 'Pareto optimality is a necessary but not sufficient condition for social welfare maximization' – justify the statement considering a $2 \times 2 \times 2$ model.
2. a) Explain the underlying assumption about the marginal utility of money for being the Hicks–Kaldor compensation be correct?
- b) Define social welfare function. How does it could be used a criteria to measure social welfare? Explain.
3. a) What are the major assumptions of social welfare maximization in a $2 \times 2 \times 2$ model.
- b) Derive the grand utility possibility frontier and determine the 'point of bliss'.
4. a) Critically explain the three key things that Pigou dealt with in analyzing social welfare.
- b) Does perfect competition ensures welfare maximization? Examine.

Section B: Information Economics

5. a) What do you mean by informational asymmetry? It is often said that as soon as the buyer of a new car drives it off of a dealer's lot, its value falls significantly below the purchase price. Why might this be true? Suppose this pattern held in 1975, but that it is no longer true today. What do you think might account for that change? Briefly Discuss.
- b) Suppose each entry-level software programmer in Palo Alto, California, has either high or low ability. All potential employers value a high-ability worker at \$12,000 per month and a low-ability worker at \$6,000. The supply of high-ability workers is $Q_H = 0.05(W - 2000)$ and the supply of low-ability workers is $Q_L = 0.1(W - 2,000)$, where W is the monthly wage. If workers' abilities are observable to employers, what are the equilibrium wages? How many workers of each type do employers hire? If workers' abilities are not observed by employers, what is the equilibrium wage? How many workers of each type do employers hire? What is the deadweight loss due to asymmetric information? Graphically present your findings.

6. a) Why might a labor market unravel? Briefly discuss.

b) What do you mean by separating and pooling equilibrium? Suppose the value of a worker's marginal product is \$55 per hour for high-ability workers and \$15 per hour for low-ability workers. The preferences of high-ability workers correspond to the utility function $U_H(E, W) = W - 4E$, and the preferences of low-ability workers correspond to the utility function $U_L(E, W) = W - 8E$. By law, everyone is required to attend at least 10 years of school. One-quarter of the population has low ability, and three-quarters have high ability. What can you say about the educational attainment of each type of worker in a separating equilibrium, and in a pooling equilibrium? Are workers better off with the most efficient separating equilibrium, or with the most efficient pooling equilibrium?

7. a) What do you mean by incentive scheme? Suppose a salesperson works for a car dealership for 40 hours per week, but may not choose to work hard all of the time. The dealership's owner cannot observe the salesperson's effort, but can observe the number of cars sold. The salesperson's personal cost of working at the dealership is $C = 1,000 + H^2$, where H is the number of hours during which he works hard. The corresponding marginal cost is $MC = 2H$. Without any effort, the salesperson will, on average, generate a profit of \$800. With each hour of high effort, he has a 10 percent chance of selling a car. Each car sale generates a profit of \$1,000. What is the efficient number of hours of high effort? How much surplus does the relationship generate? Suppose the owner gives the salesperson \$1,200 in base pay plus a bonus of \$500 for each car he sells. How hard will the salesperson work? Graphically describe an incentive scheme that leads to the efficient effort level and allows the owner to keep all of the surplus.

b) "A solution to moral hazard, albeit often an imperfect one" do you agree with this statement? Why or why not?

8. a) After reading about signaling and screening, a student named Toha complains: "The world doesn't work this way. Once an employee spends some time at a firm, her employer will learn her true ability and treat her accordingly." Do you agree with him? Why or why not?

b) In Bangladesh, bus drivers have at times been paid based on the number of riders they pick up. Discuss the likely differences in driver behavior under this plan compared to a plan in which drivers are paid by the day.

a) "Competition between employers can lead to a separating equilibrium only if low-ability workers are sufficiently numerous. If employers cannot observe each other's offers, competition cannot lead to a pooling equilibrium" do you agree with this statement? Discuss with a hypothetical example.

b) What are the four focal points that needs to be considered in the study of informational asymmetries?

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Question No. 1

(10 + 10 = 20)

- a) Define general equilibrium. In what respect it differs from the partial equilibrium. Give an example. (Handwritten: class doc)
- b) Graphically explain the problems of existence, uniqueness and stability in a general equilibrium.

Question No. 2

(10 + 10 = 20)

- a) Define externality. Examine its different types. How do externalities affect the general equilibrium solution in a $2 \times 2 \times 2$ model.
- b) Considering a $2 \times 2 \times 2$ model examine how general equilibrium solution is determined simultaneously.

Question No. 3

(10 + 10 = 20)

- a) Graphically explain how an external effect of the production affects the general equilibrium and the mechanism through which it operates.
- b) Explain the co-consumption of the two goods in a two person two goods model through Edgeworth box diagram.

Question No. 4

(10 + 10 = 20)

- a) Are all points on the contract curve of a two person two goods model Pareto efficient point of view? Why?
- b) If $MRT_{XY} = 3/2$ while $MRS_{XY} = 2$ for individual 1 and 2, should the economy produce more of X or more of Y or reach equilibrium of the two goods and services simultaneously? Why?
- c) Perfect competition leads to efficient allocation of resources in production and distribution of a particular point in time but a general equilibrium is not efficient.

5. a) Graphically examine the assumption of a $2 \times 2 \times 2$ GE model.

b) Narrate the static properties of a GE state? Explain

i) The efficiency in factor substitution in a $2 \times 2 \times 2$ GE model.

ii) The efficiency in distribution in commodities in a $2 \times 2 \times 2$ GE model.

iii) The efficiency of product mix in a $2 \times 2 \times 2$ GE model.

6. a) Show that for a simple GE model $MRPT_{x,y} = \frac{m_x}{m_y}$. Also interpret this condition through diagram.

b) Determine the prices in a simple GE model, under profit competition.